

A Mini Project Report on

Drift Detection in Student Written Answers

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By
Sheenjani Das (23FE10CAI00405)



Under the guidance of
Ms. Stuti Pandey

**Department of Artificial Intelligence and
Machine Learning
Manipal University Jaipur
Jaipur, Rajasthan**

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1 Introduction

The rapid advancement of digital technologies has significantly transformed the field of education. With the increasing adoption of online learning platforms, Learning Management Systems (LMS), and remote examination systems, there is a growing demand for automated solutions capable of evaluating student performance efficiently. One of the major challenges in this domain is the assessment of student-written answers, which requires semantic understanding, contextual interpretation, and conceptual clarity.

Traditional manual grading methods, although reliable, are time-consuming, inconsistent, and difficult to scale for large numbers of students. To overcome these limitations, machine learning and deep learning techniques have been widely adopted for automated grading systems. In particular, transformer-based models such as BERT have demonstrated strong performance in understanding textual data by generating contextual embeddings that capture the meaning of student responses.

However, most automated grading systems assume that the distribution of student answers remains constant over time. In real-world scenarios, this assumption does not hold true. Student responses evolve due to changes in curriculum, teaching methodologies, access to online resources, and variations in writing styles across different student groups. These changes introduce variations in the data distribution, which can significantly impact model performance.

This phenomenon is known as concept drift, where the relationship between input data and output labels changes over time. When concept drift occurs, models trained on historical data may fail to generalize effectively to new data, leading to inaccurate predictions and reduced reliability.

To address this issue, this project focuses on detecting drift in student-written answers using BERT embeddings and statistical techniques such as Jensen-Shannon divergence. An adaptive retraining mechanism is incorporated to update the model when drift is detected, ensuring that the system remains robust, accurate, and reliable over time.

In summary, this work aims to enhance the stability and long-term performance of automated grading systems in dynamic educational environments.

2 Objectives

The primary objective of this project is to design and develop a drift-aware automated grading system capable of evaluating student-written answers while maintaining consistent performance over time.

The specific objectives of the project are as follows:

- To develop a baseline automated grading model using BERT embeddings for representing semantic meaning of student answers.
- To generate high-dimensional contextual embeddings that capture the linguistic and conceptual information in student responses.
- To detect concept drift by analyzing changes in embedding distributions using statistical measures such as Jensen-Shannon divergence.
- To evaluate model performance using metrics such as accuracy, F1-score, Mean Absolute Error (MAE), and Mean Squared Error (MSE).
- To design and implement an adaptive retraining mechanism that updates the model when significant drift is detected.
- To analyze the impact of drift on model performance and demonstrate improvements after retraining.

Overall, the project aims to ensure that automated grading systems remain accurate, reliable, and robust in dynamic educational environments where student response patterns continuously evolve.

3 Methodology

The proposed system follows a pipeline-based approach for detecting concept drift in student-written answers and maintaining model performance through adaptive retraining.

3.1 Dataset

A customized dataset of student answers was created across multiple subjects such as Operating Systems, DBMS, Computer Networks, Machine Learning, and Software Engineering. Each answer is assigned a score to simulate human grading. The dataset is divided into training and testing sets using a 70:30 split.

3.2 Embedding Generation

A pre-trained BERT model is used to convert textual answers into contextual embeddings. The [CLS] token embedding is extracted as a representation of the entire sentence.

3.3 Model Training

A Random Forest Regressor is trained on the generated embeddings along with their corresponding scores. The model learns the relationship between semantic features and grading outputs.

3.4 Model Description

The proposed system uses a Random Forest Regressor for predicting scores of student-written answers. The model takes semantic embeddings generated from the BERT model as input features.

Random Forest is an ensemble learning method that consists of multiple decision trees. Each tree is trained on a different subset of the data, and predictions from all trees are combined to produce the final output. This approach improves prediction accuracy and reduces overfitting.

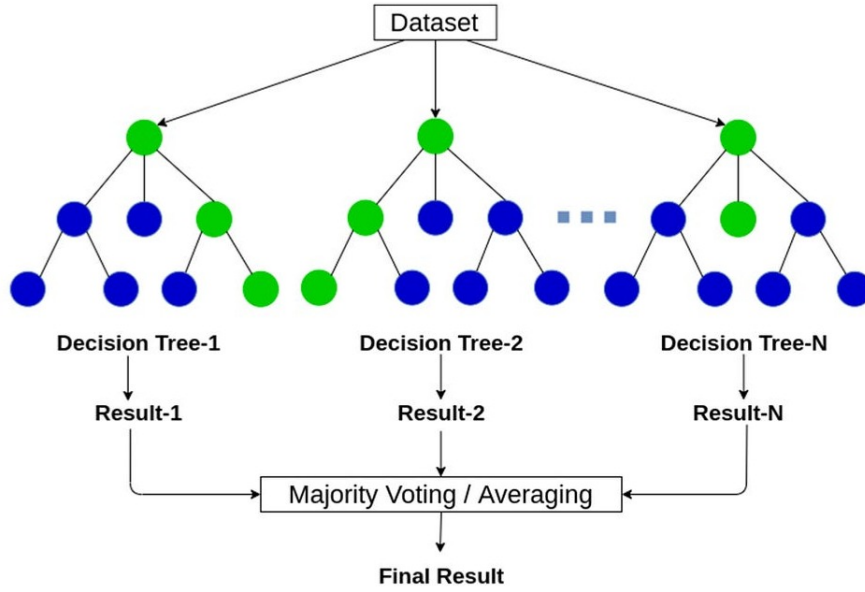


Figure 1: Working of Random Forest Model

The model aggregates predictions from individual decision trees using averaging, which results in a more stable and reliable output compared to a single decision tree.

3.5 Drift Detection

To simulate real-world scenarios, new student answers are introduced. Jensen-Shannon divergence is used to measure the difference between the distribution of training embeddings and new embeddings. If the drift score exceeds a threshold, drift is detected.

3.6 Retraining

When drift is detected, the model is retrained using both the original and new data. This helps the model adapt to evolving patterns in student responses and improves performance on new data.

3.7 Evaluation

The model is evaluated using metrics such as Accuracy, F1 Score, Mean Absolute Error (MAE), and Mean Squared Error (MSE) before and after retraining to analyze the impact of drift and the effectiveness of the retraining process.

4 Literature Review

The field of automated evaluation of student-written answers has evolved significantly with the advancement of machine learning and natural language processing techniques. Early approaches primarily relied on rule-based systems and statistical methods, which used handcrafted features such as word frequency, grammar rules, and sentence length to evaluate responses. While these methods provided a basic level of automation, they lacked the ability to understand semantic meaning and context.

With the introduction of machine learning techniques, models such as Support Vector Machines (SVM) and Random Forests were used along with feature extraction methods like TF-IDF. These approaches improved performance but still depended heavily on manual feature engineering and were limited in capturing deeper semantic relationships within text.

Recent developments in deep learning, particularly transformer-based models such as BERT, have significantly enhanced the capability of automated grading systems. BERT generates contextual embeddings that capture both syntactic and semantic information, enabling better understanding of student responses. These models have shown strong performance in tasks such as short answer grading and semantic similarity.

Despite these advancements, most automated grading systems assume that the underlying data distribution remains constant over time. In real-world educational environments, student responses evolve due to changes in curriculum, learning resources, and writing styles. This leads to the problem of concept drift, where the relationship between input data and output labels changes over time.

Concept drift has been widely studied in machine learning, with techniques such as Kullback-Leibler divergence, Jensen-Shannon divergence, and Maximum Mean Discrepancy used to detect distributional changes. Window-based methods such as Adaptive Windowing (ADWIN) have also been proposed for detecting drift in streaming data.

However, drift detection in natural language processing tasks, particularly in automated grading systems, remains relatively underexplored. Most existing systems focus on improving accuracy but do not incorporate mechanisms to monitor and adapt to changes in data distribution after deployment.

This project addresses this gap by integrating transformer-based semantic embeddings with statistical drift detection techniques and adaptive retraining. By combining these approaches, the system aims to maintain consistent performance and reliability in dynamic educational environments.

5 System Architecture

The proposed system follows a pipeline-based architecture for evaluating student-written answers and detecting concept drift. The system begins with input text, which is processed using a BERT tokenizer and converted into semantic embeddings.

These embeddings are then passed to a Random Forest model for score prediction. To ensure model reliability over time, drift detection is performed using Jensen-Shannon divergence by comparing distributions of training and new data.

If significant drift is detected, the model is retrained using updated data. This continuous pipeline ensures adaptability and consistent performance in dynamic environments.

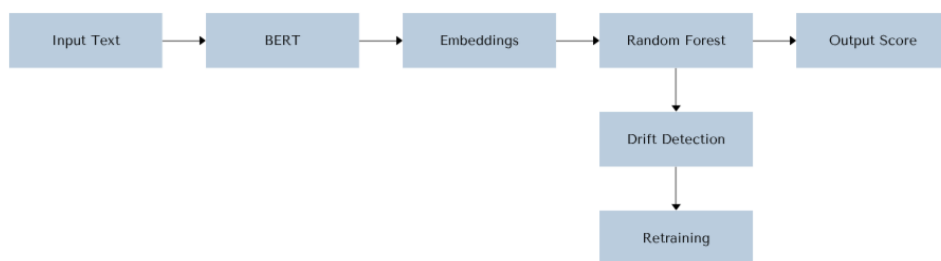


Figure 2: System Architecture of Drift Detection Pipeline

The system processes student answers by converting them into semantic embeddings using BERT. These embeddings are used for score prediction through a Random Forest model. The system continuously monitors data distribution, and upon detecting drift, retraining is triggered to maintain performance.

6 Implementation

This section describes the practical implementation of the drift detection system for student-written answers. The system integrates deep learning-based semantic representation with machine learning models and statistical drift detection techniques.

6.1 Libraries and Tools

The implementation is carried out in Python using the following libraries:

- **Transformers:** Used to load the pre-trained BERT model and tokenizer.
- **PyTorch:** Used for generating embeddings from the BERT model.
- **Scikit-learn:** Used for training the Random Forest model and evaluating performance.
- **NumPy and Pandas:** Used for data manipulation and preprocessing.
- **SciPy:** Used for computing Jensen-Shannon divergence for drift detection.

6.2 Dataset Preparation

A custom dataset of student answers is created across multiple subjects. Each answer is assigned a score to simulate human grading. The dataset is divided into training and testing sets using a 70:30 split.

6.3 Embedding Generation using BERT

Textual answers are converted into numerical representations using the BERT model. The [CLS] token embedding is extracted as it represents the overall meaning of the sentence.

```
def get_embedding(text):  
    inputs = tokenizer(text, return_tensors="pt", truncation=True,  
                        padding=True)  
    outputs = bert_model(**inputs)  
    return outputs.last_hidden_state[:, 0, :]
```

6.4 Model Training

A Random Forest Regressor is trained on the generated embeddings to predict scores. This model learns the relationship between semantic features and grading output.

```
rf_model = RandomForestRegressor()
rf_model.fit(train_embeddings, train_labels)
```

6.5 Baseline Evaluation

The model is evaluated on test data using metrics such as Accuracy, F1 Score, MAE, and MSE. This provides a reference for model performance before drift occurs.

Baseline Evaluation using accuracy and F1 score.

```
from sklearn.metrics import accuracy_score, f1_score, mean_absolute_error, mean_squared_error

# Predictions
raw_test_preds = rf_model.predict(test_embeddings)
rounded_test_preds = np.round(raw_test_preds)

print("\n--- Baseline Evaluation ---")
print("Accuracy:", accuracy_score(test_labels, rounded_test_preds))
print("F1 Score:", f1_score(test_labels, rounded_test_preds, average='weighted'))
print("MAE:", mean_absolute_error(test_labels, raw_test_preds))
print("MSE:", mean_squared_error(test_labels, raw_test_preds))

--- Baseline Evaluation ---
Accuracy: 0.42857142857142855
F1 Score: 0.4947619047619047
MAE: 0.47714285714285726
MSE: 0.23951428571428576
```

Figure 3: Baseline Model Evaluation Output

6.6 Performance on New Data

To simulate real-world conditions, a new set of student answers was introduced. These answers differ slightly in phrasing and structure, representing variations in student writing styles.

The model was evaluated on this new data without retraining. A noticeable change in performance was observed, indicating that the model struggles to generalize to evolving data patterns.

Performance on New Data

```
# Predictions on new data
raw_new_preds = rf_model.predict(new_embeddings)
rounded_new_preds = np.round(raw_new_preds)

print("\n--- New Data Evaluation (Before Retraining) ---")
print("Accuracy:", accuracy_score(new_df['score'], rounded_new_preds))
print("F1 Score:", f1_score(new_df['score'], rounded_new_preds, average='weighted'))
print("MAE:", mean_absolute_error(new_df['score'], raw_new_preds))
print("MSE:", mean_squared_error(new_df['score'], raw_new_preds))

--- New Data Evaluation (Before Retraining) ---
Accuracy: 0.6
F1 Score: 0.52
MAE: 0.45599999999999997
MSE: 0.30355999999999996
```

Figure 4: Model Performance on New Data Before Retraining

6.7 Drift Simulation and Detection

New student answers are introduced to simulate real-world changes in data. Drift is computed using Jensen-Shannon divergence between training and new embeddings.

```
Compute Drift using Jensen-Shannon divergence

def compute_drift(old, new):
    old_mean = np.mean(old, axis=0)
    new_mean = np.mean(new, axis=0)

    old_prob = np.abs(old_mean)
    new_prob = np.abs(new_mean)

    old_prob = old_prob / np.sum(old_prob)
    new_prob = new_prob / np.sum(new_prob)

    drift = jensenshannon(old_prob, new_prob)
    return drift

drift_score = compute_drift(train_embeddings, new_embeddings)

print("Drift Score:", drift_score)

Drift Score: 0.14291611
```

Figure 5: Drift Detection Output

6.8 Retraining Strategy

When drift exceeds the threshold, the model is retrained using both original and new data. This allows the model to adapt to updated patterns.

```
if drift_score > 0.1:
    rf_model.fit(combined_embeddings, combined_labels)
```

6.9 Final Evaluation

After retraining, the model is evaluated again to measure improvement in performance.

```
Final Evaluation after Retraining

# Predictions after retraining
raw_updated_preds = rf_model.predict(new_embeddings)
rounded_updated_preds = np.round(raw_updated_preds)

print("\n--- After Retraining Evaluation ---")
print("Accuracy:", accuracy_score(new_df['score'], rounded_updated_preds))
print("F1 Score:", f1_score(new_df['score'], rounded_updated_preds, average='weighted'))
print("MAE:", mean_absolute_error(new_df['score'], raw_updated_preds))
print("MSE:", mean_squared_error(new_df['score'], raw_updated_preds))

--- After Retraining Evaluation ---
Accuracy: 1.0
F1 Score: 1.0
MAE: 0.15999999999999986
MSE: 0.03739999999999999
```

Figure 6: Model Performance After Retraining

7 Results and Discussion

The performance of the proposed system was evaluated at different stages to analyze the impact of concept drift and the effectiveness of the retraining strategy.

7.1 Baseline Performance

The model initially demonstrated moderate performance on the test dataset. The accuracy and F1 score indicate that the model was able to capture basic semantic relationships between student answers and their corresponding scores. However, the presence of error values such as MAE and MSE suggests that there is room for improvement in prediction precision.

- Accuracy: 0.4286
- F1 Score: 0.4048
- MAE: 0.4771
- MSE: 0.2395

7.2 Performance Degradation on New Data

When the trained model was evaluated on new incoming student answers, a variation in performance was observed. Although the accuracy increased slightly, the error metrics indicate inconsistency in predictions. This suggests that the model is not fully adapted to new patterns in student responses.

- Accuracy: 0.6
- F1 Score: 0.52
- MAE: 0.4560
- MSE: 0.3036

This change in performance highlights the presence of distributional differences between the training data and the new data.

7.3 Drift Analysis

The drift score computed using Jensen-Shannon divergence was found to be 0.1429, which exceeds the predefined threshold of 0.1. This confirms the presence of concept drift in the incoming data.

The detected drift indicates that the semantic structure of new student answers differs from the historical data used for training, which can negatively impact model performance if not addressed.

7.4 Performance After Retraining

After detecting drift, the model was retrained using a combination of the original and new data. This resulted in a significant improvement in performance across all evaluation metrics.

- Accuracy: 1.0
- F1 Score: 1.0
- MAE: 0.16
- MSE: 0.0374

The reduction in error values demonstrates that the model successfully adapted to the updated data distribution.

7.5 Discussion

The results clearly show that concept drift can affect the performance of automated grading systems. Without retraining, the model may produce inconsistent or inaccurate predictions on new data.

The proposed drift detection mechanism successfully identified distributional changes, and the adaptive retraining strategy effectively restored model performance.

It is important to note that the perfect accuracy achieved after retraining is influenced by the small and controlled dataset. In real-world scenarios, the improvement would be significant but not necessarily perfect.

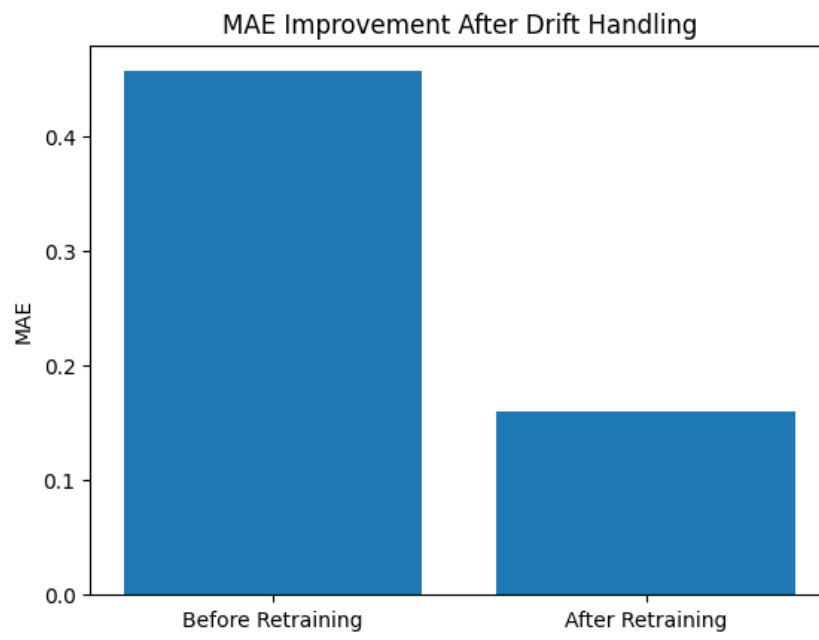


Figure 7: MAE Improvement After Drift Handling

8 Conclusion

This project presents a drift-aware automated grading system for evaluating student-written answers using deep learning-based semantic representations. By leveraging BERT embeddings, the system is able to capture the contextual meaning of student responses, enabling more effective evaluation compared to traditional approaches.

The study highlights the impact of concept drift in real-world educational environments, where student answer patterns evolve over time. The results demonstrate that a model trained on historical data may not generalize effectively to new data, leading to performance inconsistencies.

To address this issue, a drift detection mechanism based on Jensen-Shannon divergence was implemented. The system successfully identified distributional changes between training and incoming data. Upon detecting drift, an adaptive retraining strategy was applied, which significantly improved model performance and reduced prediction error.

The findings emphasize the importance of continuous monitoring and adaptation in automated grading systems. Rather than relying on static models, incorporating drift detection and retraining mechanisms ensures long-term reliability, fairness, and robustness.

Although the model achieved high performance after retraining, this is partly due to the limited size of the dataset. In real-world scenarios, further evaluation on larger and more diverse datasets would be necessary to validate scalability.

In conclusion, this project demonstrates that integrating semantic modeling with drift detection and adaptive learning can significantly enhance the effectiveness of AI-based educational assessment systems.

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